Does a Flood Map Transfer to a New Flood?

Code

Five water-mapping methods, eleven floods, and what changes from one flood to the next

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PUBLISHED
October 8, 2026

Interactive application

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1 Introduction

Floods arrive with storm clouds, which blind ordinary (optical) satellites. **Synthetic-aperture radar** isn't blind: Sentinel-1 sends its own microwave pulse, day or night, through cloud. Calm water reflects that pulse away from the satellite like a mirror, so it appears **dark**; rough land scatters some back.

That makes radar the workhorse of flood mapping — but every flood is different: different landscapes, seasons, water depths, wind. A method tuned on past floods is always applied to a new one. This project measures how much that matters, comparing five methods of increasing complexity on 11 floods from six continents, each scored on a flood it has never seen.

2 Data

Sen1Floods11 (Bonafilia et al., CVPR Workshops 2020): 446 hand-labelled 512×512-pixel chips (~10 m pixels) from 11 flood events — Bolivia, Ghana, India, Mekong, Nigeria, Pakistan, Paraguay, Somalia, Spain, Sri Lanka, USA. Each chip has Sentinel-1 radar (VV and VH, in dB), Sentinel-2 optical bands from the same period (often partly cloudy), a hand label (water / not water), and the JRC permanent-water mask, which separates rivers and lakes from new floodwater.

3 Methods

Method	What it uses
VV threshold	Water where VV is below one dB value, fit on the training floods
Authors' Otsu	The dataset authors' per-scene Otsu threshold — adapts to each scene, no training
LightGBM	Gradient-boosted trees on each pixel's VV, VH, their ratio, and neighbourhood mean / spread at 50 m and 150 m
U-Net (radar)	Convolutional segmentation network (ResNet-34 encoder) on VV, VH and their ratio
U-Net (radar + optical)	The same network with six Sentinel-2 bands and the NDWI water index added

Two evaluations:

1. **Official split** — the dataset's own train / validation / test chips (drawn from the same 10 floods) plus Bolivia, held out entirely.
2. **Leave-one-event-out** — for each of the 11 floods: train on the other 10, score on that one. The U-Net chooses its best epoch on 10% of the *training* chips, never the held-out flood.

Metric: **water IoU** (intersection over union), pooled over all valid pixels.

4 Results

4.1 On familiar floods, complexity pays

	test	Bolivia (unseen)
VV threshold	0.464	0.441
Authors' Otsu	0.554	0.718
LightGBM (radar + texture)	0.637	0.405
U-Net (radar)	0.672	0.597
U-Net (radar + optical)	0.830	0.811

On the official test split — chips from floods the models trained on — each step up in complexity helps, and adding optical bands helps most.

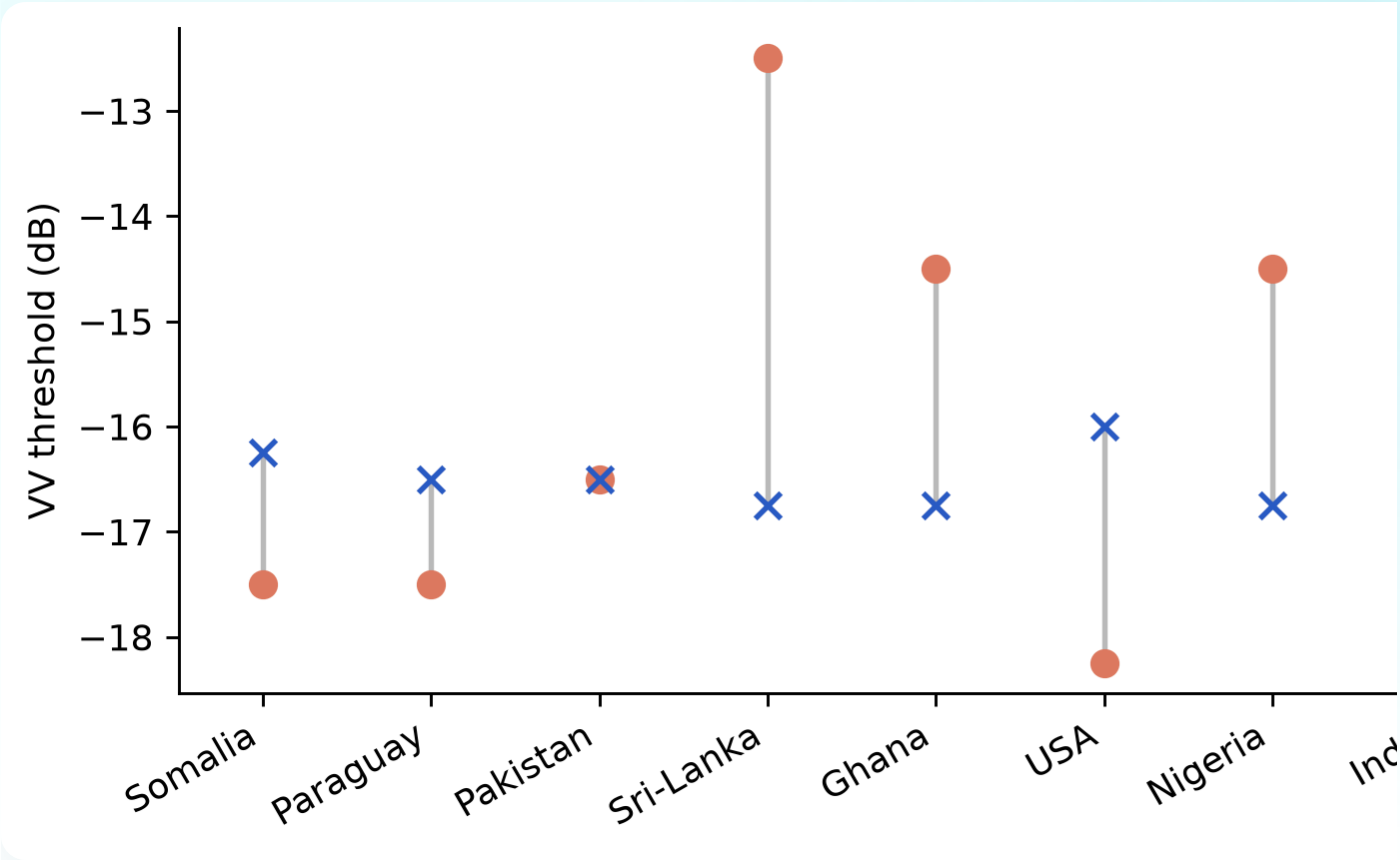
4.2 On a new flood, the picture changes

event	Threshold	Otsu	LightGBM	U-Net radar	U-Net radar+optical
Somalia	0.15	0.15	0.22	0.27	0.70
Paraguay	0.43	0.45	0.55	0.53	0.60
Pakistan	0.24	0.15	0.30	0.32	0.65
Sri-Lanka	0.16	0.65	0.59	0.68	0.85
Ghana	0.35	0.41	0.39	0.49	0.68
USA	0.29	0.51	0.51	0.50	0.78
Nigeria	0.55	0.64	0.65	0.76	0.89
India	0.47	0.53	0.51	0.53	0.71
Spain	0.63	0.67	0.72	0.56	0.85
Bolivia	0.44	0.72	0.43	0.68	0.81
Mekong	0.77	0.78	0.83	0.84	0.92
Pooled	0.43	0.50	0.57	0.59	0.78

Pooled over all 11 held-out floods, **U-Net (radar + optical)** scores best (IoU 0.78). Compared with the official test split (familiar floods): VV threshold drops 0.03, Authors' Otsu drops 0.06, LightGBM (radar + texture) drops 0.07, U-Net (radar) drops 0.08, U-Net (radar + optical) drops 0.05. The radar + optical U-Net is the best method on 11 of 11 held-out floods.

4.3 Why fixed rules break: the right threshold moves

▶ Code

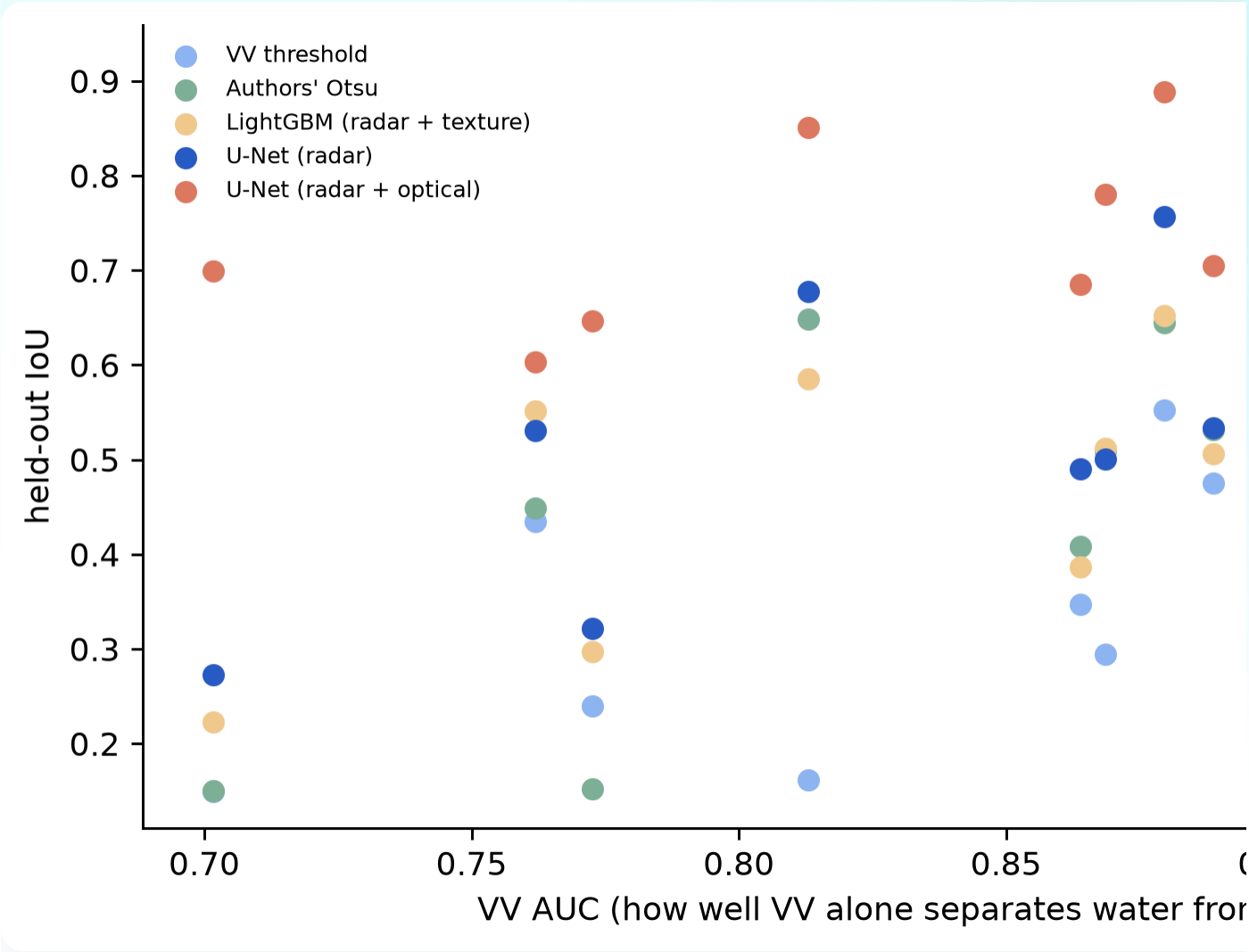


Each flood's own best VV threshold (dots) against the threshold learned from the other ten floods (crosses).

Each flood's own best threshold ranges from **-18.25 to -12.50 dB** (5.8 dB). Any threshold learned from the other floods lands between -16.75 and -16.00 dB — essentially one number. The worst mismatch is **Sri-Lanka**: its best threshold is -12.50 dB against a learned -16.75 dB, and the fixed threshold scores IoU 0.16 there while per-scene Otsu scores 0.65.

4.4 For radar alone, the flood matters more than the method

▶ Code



Held-out IoU against how separable water is in that flood's radar (VV AUC).

How visible water is to radar varies enormously: in **Somalia** water is only 2.6 dB darker than land (AUC 0.70); in **Mekong**, 12.0 dB (AUC 0.97). Every method's score tracks this closely (correlations VV threshold 0.77, Authors' Otsu 0.84, LightGBM (radar + texture) 0.68, U-Net (radar) 0.75, U-Net (radar + optical) 0.68).

For the radar-only methods, a method's IoU varies by **0.61** on average from one flood to another, while different methods on the same flood differ by only **0.19**: which flood you're mapping predicts the result better than which radar method you use. Adding optical bands changes that: the radar + optical U-Net's scores span only 0.32 across floods, and its biggest gains come exactly on the floods where radar struggles most (e.g. Somalia: radar-only best 0.27, radar + optical 0.70).

4.5 Where water is missed

Recall (share of labelled water found) on held-out floods, for permanent water (rivers, lakes) and new floodwater:

	new floodwater	permanent water
VV threshold	0.45	0.75
Authors' Otsu	0.60	0.99
LightGBM (radar + texture)	0.52	0.94
U-Net (radar)	0.62	0.91
U-Net (radar + optical)	0.81	0.99

Split by vegetation (NDVI from clear-sky Sentinel-2 pixels), the best model's recall collapses as vegetation increases — water among or under plants is close to invisible. Labelled water is rare in vegetated pixels, so these are small samples:

ndvi	water share	recall	false-alarm rate
bare / water (< 0.2)	0.639	0.91	0.0845
sparse (0.2-0.4)	0.019	0.24	0.0283
moderate (0.4-0.6)	0.004	0.05	0.0021
dense (> 0.6)	0.002	0.01	0.0002

5 Limitations

- **Eleven floods is a small sample** for statements about "a new flood"; per-flood scores rest on 15 to 69 chips each, and some floods have little water.
- **Optical data needs clear skies.** The radar + optical model was trained and tested on chips whose Sentinel-2 image is sometimes partly cloudy, but how it behaves under complete cloud cover wasn't tested separately — and real flood response often has no cloud-free optical image at all.
- **Hand labels have their own uncertainty**, especially at shorelines and in flooded vegetation.
- **One processing chain.** All radar here was processed identically; mixing radar products from different pipelines can shift values by several dB and break models silently.

6 Conclusion

On familiar floods, more complex models win clearly. On a new flood, every method loses ground, and for radar-only methods the result depends more on the flood than on the method: the

brightness of water in radar shifts by several dB between floods, so any single learned threshold is wrong somewhere, and a per-scene method like Otsu beats it exactly on those floods. Among radar-only methods, **U-Net (radar)** transfers best (pooled IoU 0.59). Adding optical bands gives the strongest model overall (pooled IoU 0.78) — with the caveat that real flood response often has no cloud-free optical image, when only the radar results apply.